***Goal representation:*** The cognitive module clearly defines the AI system's goals. These could be simple and explicit such as "move to a certain location," or complex and implicit, such as "maximise operational efficiency." These lay the foundation for all the subsequent actions that the AI system will take.

***Planning:*** The agent's planning module generates a structured approach to achieve the goals in mind. This may include creating a set of steps that are to be followed in a specific order or breaking down the main goal into smaller, more manageable sub-goals.

***Decision-making:*** The choice-making or decision-making module decides on what actions can be taken, checks the environment's current situation, monitors what plan it is executing, and observes what objectives it desires to achieve. It then chooses the one that has the highest likelihood of yielding the desired result. For example, an AI system for a drone helps it take a detour around obstacles and alter its course to the target.

***Action execution:*** Once a decision has been made, the action module executes it. This action can be physical like lifting or pressing something, or can be virtual such as sending signals, reconfiguring, or even buying things online.

***Learning:*** The learning module of the system enables it to learn with time and improve its performance. It learns to change its strategies based on feedback from the environment, probably using reinforcement learning or supervised learning. This form of continuous learning enhances accuracy as well as efficiency for the agentic AI system.

## Comparing agentic AI with other AI systems

Agentic AI is defined by its ability to act on its own, make decisions on its own, and adapt to new data or environmental changes. A great example would be an autonomous car, which decides automatically on routes and adjustments based on traffic, weather, or roadblocks without human intervention.

Generative AI is all about creating new content. Unlike data analysis, it learns patterns and uses that to create new examples such as generating text (GPT-4) or creating images (DALL-E). This kind of AI focuses more on creativity and replication of real-world results based on the learned patterns.

Classic machine learning is linked to pattern identification and forecasting. This can be better comprehended using models that analyse past data and decide based on learned patterns. Classic ML use cases are more common in fraud detection, where past data is used to train a model to detect fraud.

Table 1 summarises the main differences between these AI systems.

## Who is investing in agentic AI?

***AutoGPT by OpenAI****:* This open source project allows large language models to do tasks without human input -- for example, market research or business planning.

***AlphaStar and AlphaGo (DeepMind):*** These rely on reinforcement learning for creating strategic AI agents and have the ability to make real-time moves in games like StarCraft II and Go.

***Meta's CICERO (Diplomacy AI):*** CICERO is capable of strategic negotiation and cooperation.

***Tesla's full self-driving (FSD) AI****:* This software enables a real world experience by self-driving cars. Agentic AI enables continuous learning from data gathered through actual experiences.

***Google's PaLM-SayCan for robotics:*** This effort integrates Google's PaLM language model with robots to enable them to understand and execute complex instructions. It is a step towards robots that are more sensitive to human instructions in dynamic environments.

## Single-agent vs multi-agent AI systems

In a single-agent system, the architecture is focused on one AI agent that uses a set of tools to address some problems. This includes perception, reasoning, action, and learning, as described earlier.

A multi-agent system (MAS) architecture has a number of autonomous agents working together to resolve problems of a larger magnitude. Here's how it works.

***Multiple AI agents:*** Each independent agent, though in the

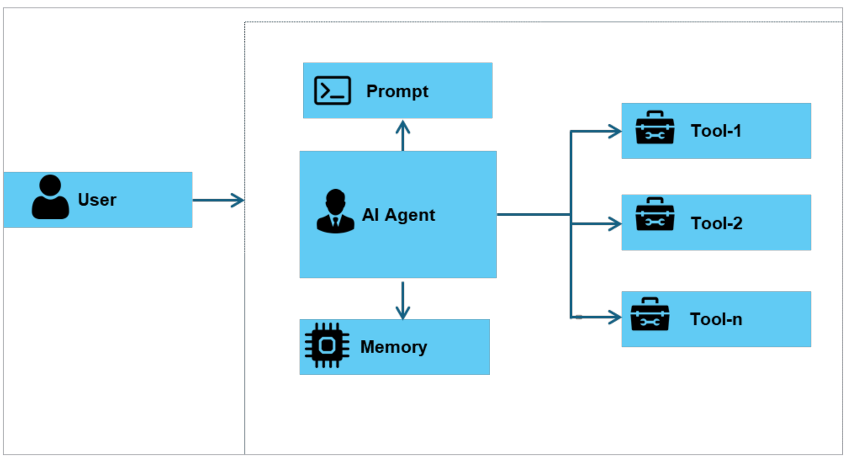


Figure 5: Single-agent system